Ethan Descamps

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Education

University of California, Berkeley August 2022 - December 2026

B.S. Mechanical Engineering

- **Activities:** Formula Electric at Berkeley **(FEB)**, Tau Beta Pi Engineering Honors Society **(TBP)**
- **Relevant Coursework:** Engineering Data Analysis, Structure and Interpretation of Computer Programs, Mechanical Behavior of Engineering Materials, Solid Mechanics, Thermodynamics, Heat Transfer, Rigid Body Dynamics, Data Science for Engineers

Alma Mater Studiorum – Università di Bologna Fall 2025

B.S. Mechanical Engineering

- **Master's Level Courswork:** Virtual Vehicle Design and Product Reliability

Experience

Abbott Cardiac Rythm Management Jan 2026 - June 2026

Mechanical Research and Development Engineering Co-op *Sylmar, CA*

- Led product development for MRI-compatible implantable device solutions, generating **50+ design iterations** through rapid prototyping and 3D printing while applying human-factors engineering and ergonomic design principles; refined concepts using direct feedback from clinicians, Human Factors specialists, and cross-functional engineering teams during clinical evaluation
- Utilized Nikon XT H 225 ST X-ray CT scanning to reverse engineer the FreeStyle Libre 3, recreating the full assembly, internal components, and insertion mechanism from scratch in CAD to support product architecture analysis and future development
- Designed go-no-go gauges using GD&T principles to ensure accurate dimensional checks and repeatable inspection for a next-generation leadless pacemaker
- Drove DFM improvements through change orders across multiple products, reducing piece-part count and error-proofing production and documentation workflows for tooling and materials release
- Converted 2D electrical schematics and PCB layouts into detailed 3D CAD models of a Bluetooth-enabled pacemaker system, enabling packaging studies, design reviews, and cross-functional integration between electrical and mechanical teams

Pankl Racing Systems June 2025 - August 2025

Research and Development Engineering Intern *Styria, Austria*

- Optimized Formula One 2026 single cylinder test bench engine through intensive fatigue life part analyses
- Applied NX Siemens for design modifications, ANSYS Workbench (steady-state thermal and static structural) with external CFD data for load modeling, and FEMFAT for fatigue analysis, leading to targeted upgrades specifically within the engine liner and water jacket resulting in a crucial **19.83%** weight reduction and refined heat dissipation

FSAE Electric Berkeley Sep 2023 - May 2025

Design and Aerodynamics Engineer *Berkeley, CA*

- Designed and optimized the entire front wing mounting structure utilizing SolidWorks and ANSYS software; conducted comprehensive iterative finite element analyses to refine stress distribution, and applied advanced topology optimization techniques to enhance structural integrity and performance while reducing material usage
- Achieved a **20%** reduction in weight for the front wing design, successfully maintaining structural rigidity and ensuring a robust Factor of Safety, demonstrating efficient and innovative engineering solutions

Jacobs Hall of Design and Innovation Feb 2024 - May 2025

Student Supervisor *Berkeley, CA*

- Maintained makerspace equipment and tools, and assist **200+** regular users with safe operation
- Facilitated and aided makerspace users with the use of hand/power tools, CNC mill, Universal and FabLight laser cutters, waterjet cutter, and FDM, PolyJet, and SLA industrial 3D printers

Projects

Automatic Cat Feeder Dec 2024

- Designed an automatic cat feeder prototype using Onshape, integrating advanced sensors (IMU, ultrasonic, load cell) and actuators, using ESP32 microcontrollers to automate food dispensing and monitor supply levels

Hands On PCB Engineering Nov 2024

- Created and manufactured a custom PCB design for a vintage record player
- Cut down the number of MCUs needed by **50%** to perform the same functions of a double tone-armed turned table while simultaneously condensing unorganized wires and breadboards into a single hand-soldered PCB

Skills

Engineering: CAD, FEA, Topology Optimization, CFD, Rapid Prototyping, DFM, ECAD, 3D Printing, Laser Cutting, CNC, GD&T
Technology: Creo PTC, Solidworks, Ansys Workbench, Siemens NX, Python, Matlab, KiCad, Fusion 360, Rhino, SQL, Scheme, Onshape
Languages: English (Fluent), French (Fluent), Spanish (Limited Working Proficiency)